Research routes toward recursive self-improvement

Selected milestones by first public year | Historical roots and the foundation-model era

Data & training	Tasks & curricula	Memory & skills	Code & workflows	Feedback & judges	Meta-level improvement
Synthetic data Persistent updates	Task generation Environment feedback	Reflective memory Reusable knowledge	Prompts and code Agent architectures	Learned rewards Evaluator learning	Update strategies Research automation

2026

- MetaClaw (Memory & skills)
- Prime Agent (Code & workflows)
- Meta^n (Meta-level improvement)
- Ind. Bilevel Autoresearch (Meta-level improvement)
- Hyperagents (Meta-level improvement)

2025

- SEAL (Data & training)
- AgentEvolver (Tasks & curricula)
- ACE (Memory & skills)
- GEPA (Code & workflows)
- AlphaEvolve (Meta-level improvement)
- R-Zero (Tasks & curricula)
- Dynamic Cheatsheet (Memory & skills)
- DGM (Code & workflows)
- Process Self-Rewarding (Feedback & judges)
- TTRL (Data & training)
- Absolute Zero (Tasks & curricula)
- A-MEM (Memory & skills)
- SICA (Code & workflows)
- AI Scientist-v2 (Meta-level improvement)

2024

- SCoRe (Data & training)
- RISE (Data & training)
- Quiet-STaR (Data & training)
- SPIN (Data & training)
- AgentGym (Tasks & curricula)
- AFlow (Code & workflows)
- CREAM (Feedback & judges)
- Self-Taught Evaluators (Feedback & judges)
- Meta-Rewarding (Feedback & judges)
- Self-Rewarding (Feedback & judges)
- ADAS (Code & workflows)
- TextGrad (Code & workflows)
- AI Scientist (Meta-level improvement)

2023

- ReST^{EM} (Data & training)
- ReST (Data & training)
- ExpeL (Memory & skills)
- Voyager (Memory & skills)
- Reflexion (Memory & skills)
- DSPy (Code & workflows)
- OPRO (Code & workflows)
- CRITIC (Feedback & judges)
- STOP (Meta-level improvement)
- Promptbreeder (Meta-level improvement)

2022

- Self-Instruct (Data & training)
- STaR (Data & training)
- Constitutional AI (Feedback & judges)

Historical roots

- **Gödel Machine** — 2003 | Provable self-modification
- **Learned optimizers** — 2016 | Learning update rules
- **POET** — 2019 | Tasks and solvers co-evolve

Marks: selected affiliations | Ind.: independent researchers | Branches: research routes, not technical ancestry